Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	11
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1994.91	1519	1026.72	px^2
major_axis_length_px	lacuna shape	85.7455	86.1566	22.5145	px
minor_axis_length_px	lacuna shape	30.4546	26.8577	8.8339	px
aspect_ratio	lacuna shape	2.9668	3.0363	0.8704	unitless
eccentricity	lacuna shape	0.92	0.9442	0.0643	unitless
solidity	lacuna shape	0.8555	0.8762	0.0782	unitless
roots_count	headline	5.9091	6	2.0715	count
ring_length_r30_px	headline	315.182	334	82.5843	px
ring_length_r60_px	reported	864.182	883	157.884	px
owned_length_px	ownership-dependent	2076.51	1597.87	1461.62	px
edge_count	ownership-dependent	97.3636	67	70.6049	count
mean_edge_length_px	ownership-dependent	22.0665	21.1756	2.5685	px
perimeter_px	normalised	221.572	210.244	51.1926	px
ring_area_r30_px2	normalised	8909.82	8646	1432.67	px^2
ring_area_r60_px2	normalised	22488.8	22056	3166.3	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0358072	0.0352764	0.0108291	px^-1
ring_density_r60	normalised	0.0388108	0.0388481	0.00773588	px^-1
roots_per_100px_perimeter	normalised	2.7497	2.6998	1.0704	per 100 px
ring_attached_length_r30_px	network v2	256.909	249	94.1652	px
ring_attached_length_r60_px	network v2	596.455	591	215.57	px
ring_length_w_r30_px	network v2	356.286	372.078	92.2592	px
ring_length_w_r60_px	network v2	975.732	994.323	173.348	px
sholl_crossings_r10	network v2	7.5455	7	2.2523	count
sholl_crossings_r20	network v2	10.7273	12	2.3703	count
sholl_crossings_r30	network v2	12.7273	12	3.5239	count
ring_width_mean_r30_px	width	7.1227	7.0394	0.4395	px

Per field
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measure	value	units
analysed_area_px2	1.02271e+06	px^2
total_skeleton_length_px	36044	px
canalicular_length_density_per_px	0.0352435	px^-1
junction_count	804	count
junction_density_per_px2	0.00078615	px^-2
lacuna_count	14	count
lacunae_per_px2	1.369e-05	px^-2
skeleton_component_count	272	count

mean_component_length_px	132.515	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0398811	px^-1
field_density_without_flagged_per_px	0.0361495	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	560	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0352435	px^-1
field_length_density_with_unattached_per_px	0.0357911	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7083	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	34694	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	510	36.0388	20.5481	1.7539	0.8215	0.8095	-0.2657	13.3863	106.873	True
2	448	43.7126	19.8458	2.2026	0.891	0.6767	1.435	7.0045	363.067	True
3	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
4	1062	83.8665	19.2697	4.3522	0.9732	0.6923	-1.1353	135.124	920.395	False
5	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
6	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
7	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
8	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
9	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
10	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
11	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
12	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
13	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
14	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False

Network per lacuna

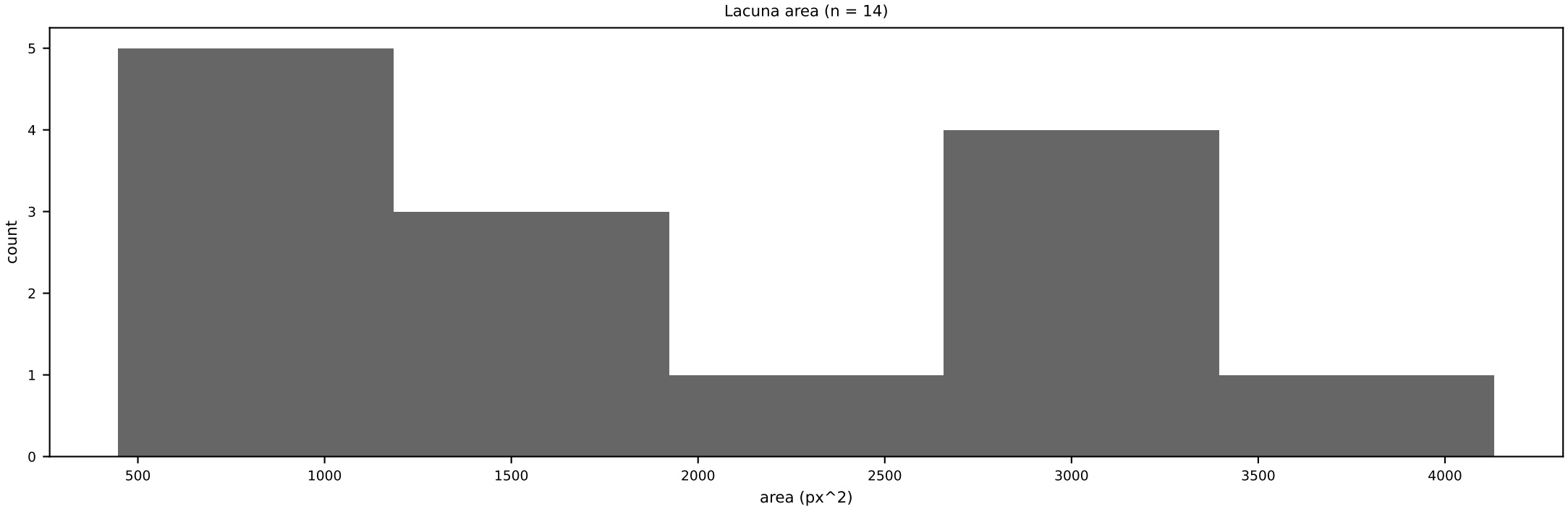
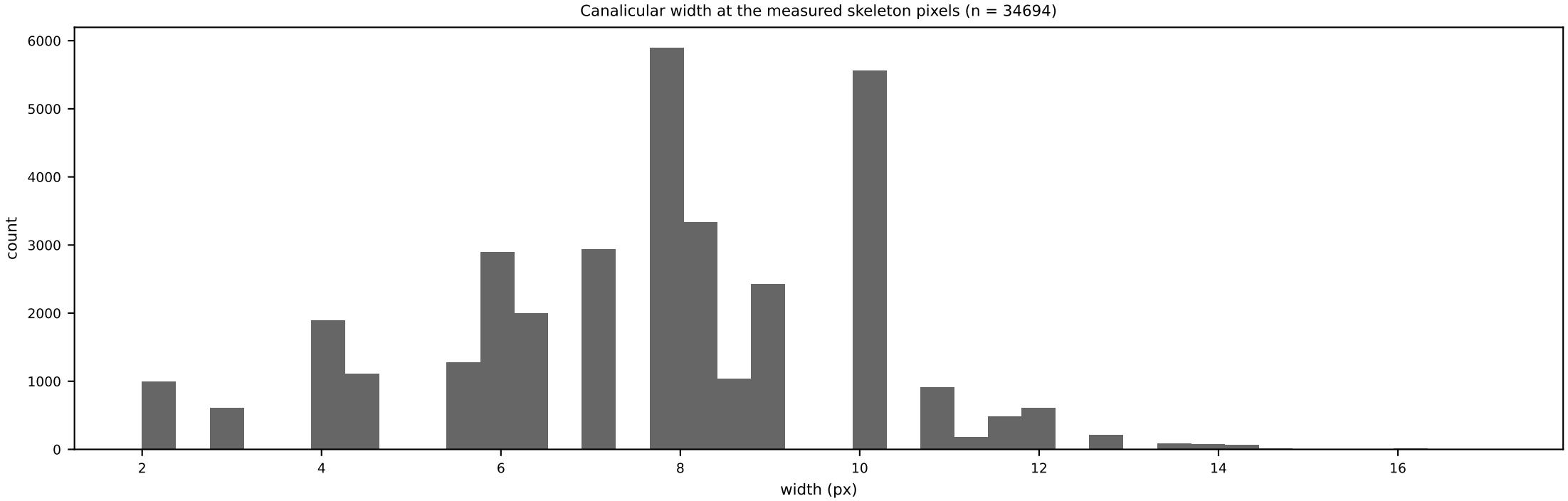
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	96	220	3081.49	141	21.8546	
2	3	42	91	36.3137	4	9.0784	
3	8	329	752	1973.27	97	20.343	6.7412
4	10	375	1001	4443.21	216	20.5704	6.9373
5	5	334	904	3405.89	164	20.7676	7.2307
6	3	206	685	1275.64	60	21.2607	7.8564
7	6	352	883	4044.54	191	21.1756	7.3054
8	7	463	976	3223.29	144	22.384	7.4331
9	8	299	760	711.617	30	23.7206	7.398
10	5	350	1143	1597.87	65	24.5826	7.7674

11	5	233	629	362.463	13	27.8818	6.5248
12	6	305	862	1346.99	67	20.1043	6.8676
13	3	174	700	610.529	25	24.4212	7.0394
14	7	346	971	557.961	29	19.24	6.646

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	105.154	3701	10161	0.6475	0.597	0.0259389	0.0216514	1.902
2	132.847	3699	7686	0.6031	0.5656	0.0113544	0.0118397	2.2582
3	219.681	8976	20715	1	0.9689	0.0366533	0.0363022	3.6416
4	210.244	8511	22591	1	1	0.0440606	0.0443097	4.7564
5	335.421	12178	29932	1	1	0.0274265	0.0302018	1.4907
6	167.238	7485	20582	1	1	0.0275217	0.0332815	1.7939
7	243.522	9472	22056	1	1	0.0371622	0.0400345	2.4638
8	167.924	7349	17555	1	1	0.0630018	0.0555967	4.1686
9	282.894	8678	19332	0.813	0.7462	0.0344549	0.0393131	2.8279
10	263.48	10172	24552	1	1	0.0344082	0.0465542	1.8977
11	174.51	7570	20770	1	1	0.0307794	0.0302841	2.8652
12	210.167	8646	22189	1	1	0.0352764	0.0388481	2.8549
13	185.823	7980	21610	1	1	0.0218045	0.0323924	1.6144
14	259.279	9669	24825	1	1	0.0357845	0.0391138	2.6998

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	81	151	103.426	241.823	3	2	4
2	38	38	43.1421	98.7696	1	0	0
3	190	462	366.907	831.955	7	12	14
4	323	779	430.747	1149.46	10	12	12
5	307	587	372.078	1001.77	8	11	9
6	166	325	236.137	787.625	6	7	7
7	281	591	392.907	994.323	9	13	12
8	451	949	523.576	1094.49	11	13	15
9	180	495	336.25	850.855	6	11	11
10	248	848	392.907	1281.35	6	13	19
11	149	293	264.38	728.139	4	10	10
12	249	610	353.019	973.222	7	8	17
13	137	390	199.167	799.453	5	7	11
14	325	727	387.321	1091.26	10	12	14

682_z23: distributions, pre-validation, px